OSeMOSYS UPME Model Description

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Executive Summary

This repository runs a Pyomo implementation of OSeMOSYS, a long-term energy-system optimization model. The current model instance is best described as a large-scale deterministic linear programming problem, solved through Pyomo using HiGHS, Gurobi, or GLPK.

Strictly, the code can represent a mixed-integer linear program because it declares integer variables for technology unit investments. However, in the current CSV dataset those discrete unit-size constraints are inactive because `CapacityOfOneTechnologyUnit` is zero everywhere. In practical terms, the instance being solved behaves as a linear program.

Mathematical Problem Type

The model minimizes total discounted system cost subject to technical, economic, demand, emissions, reserve-margin, and policy constraints.

Objective:

```
Minimize total discounted system cost over all regions and years
```

The cost accounting includes investment costs, fixed and variable operating costs, emissions penalties, salvage value, disposal cost, and recovery value.

Solver Interfaces

The repository maps solver names to Pyomo solver factories as follows:

Requested solver	Pyomo solver factory
`highs`	`appsi_highs`
`gurobi`	`gurobi`
`glpk`	`glpk`

If `GLPX` is mentioned, the intended solver is almost certainly `GLPK`.

Current Model Instance Size

The following counts were obtained by instantiating the Pyomo model from the repository's current `CSV` input folder.

Characteristic	Count
Regions	1
Technologies	394
Fuels	109
Emissions	3
Years	33
Timeslices	1
Modes of operation	1
Active storage technologies	0
UDC groups	1
Pyomo variable entries	364,717
Declared integer variable entries	13,002
Binary variables	0
Active constraints/equations	409,207
Approximate nonzero variable appearances in constraints	4,736,264

Important note: storage feature files exist, but `STORAGE.csv` has no active values. Therefore, storage variables and storage equations instantiate to zero active entries in the current run.

Main Decision and Accounting Variables

Variable family	Meaning
`NewCapacity`	New technology capacity installed
`RateOfActivity`	Technology activity or dispatch
`CapitalInvestment`	Investment cost accounting
`OperatingCost`	Fixed and variable operating cost accounting
`AnnualTechnologyEmission`	Emissions by technology, emission type, and year
`AnnualEmissions`	Total annual emissions by emission type
`TotalDiscountedCost`	Annual discounted system cost
`TotalCapacityInReserveMargin`	Capacity counted toward reserve margin
`NumberOfNewTechnologyUnits`	Integer unit count, declared but inactive in this dataset

Largest Active Variable Blocks

Variable block	Count
`AnnualTechnologyEmissionByMode`	39,006
`AnnualTechnologyEmission`	39,006
`AnnualTechnologyEmissionPenaltyByEmission`	39,006
`NewCapacity`	13,002
`RateOfActivity`	13,002
`CapitalInvestment`	13,002
`OperatingCost`	13,002
`TotalDiscountedCostByTechnology`	13,002
`NumberOfNewTechnologyUnits`	13,002

Main Constraint Categories

The model includes the following main groups of constraints:

Category	Purpose
Demand balance	Energy supplied must satisfy demand by fuel, year, and timeslice
Capacity constraints	Activity cannot exceed available installed capacity
Investment constraints	New capacity is bounded by annual min/max investment limits
Activity constraints	Technologies may have lower or upper activity bounds
Emissions accounting and limits	Emissions are calculated, penalized, and optionally bounded
Reserve margin	Capacity adequacy constraints for selected technologies and fuels
User-defined constraints	Additional policy or system constraints through UDC parameters
Storage constraints	Supported by the code, but inactive in the current dataset

Largest Active Constraint Blocks

Constraint family	Active constraints
`AnnualEmissionProductionByMode`	39,006
`AnnualEmissionProduction`	39,006
`EmissionPenaltyByTechAndEmission`	39,006
`ConstraintCapacity`	13,002

Constraint family	Active constraints
`PlannedMaintenance`	13,002
`OperatingCostsVariable`	13,002
`OperatingCostsFixedAnnual`	13,002
`TotalAnnualMinCapacityConstraint`	13,002
`TotalAnnualMinNewCapacityConstraint`	13,002
`TotalAnnualTechnologyActivityLowerlimit`	13,002
`EnergyBalanceEachTS5`	3,597
`EnergyBalanceEachYear4`	3,597

In optimization terminology, the word "equations" is often used informally for all expanded active constraints, including both equalities and inequalities. The current model has 409,207 active constraints/equations.

Technologies With Restrictions

There are 394 technologies in the active model set. Using non-default capacity, activity, reserve-margin, and UDC parameters as the definition of having an active restriction, 343 technologies have at least one active restriction.

Restriction parameter	Technologies affected	Active rows
`TotalAnnualMaxCapacity`	11	363
`TotalAnnualMaxCapacityInvestment`	312	10,296
`TotalAnnualMinCapacityInvestment`	10	49
`TotalTechnologyAnnualActivityUpperLimit`	8	234
`TotalTechnologyAnnualActivityLowerLimit`	153	4,219
`TotalTechnologyModelPeriodActivityUpperLimit`	5	5
`ReserveMarginTagTechnology`	19	627
`UDCMultiplierTotalCapacity`	20	660

No active technology restrictions were found for the following categories in the current dataset:

Parameter or category	Status
`CapacityOfOneTechnologyUnit`	All zero, so integer unit constraints are inactive
`RETagTechnology`	No active technology tags
Technology activity by mode limits	No nonzero active values
Storage technology constraints	Storage set is empty

Suggested Narrative Description

The model can be described as follows:

> We run a Pyomo implementation of OSeMOSYS, a deterministic long-term energy-system optimization model. The model minimizes total discounted system cost while satisfying demand, capacity, activity, emissions, reserve-margin, and user-defined policy constraints. The current instance covers 1 region, 394 technologies, 109 fuels, 3 emissions, and 33 model years. It expands to 364,717 Pyomo variable entries and 409,207 active constraints. Although the formulation supports integer capacity-unit decisions, the current dataset has no active unit-size constraints, so the solved problem is effectively a large linear program. The model can be solved with HiGHS, Gurobi, or GLPK through Pyomo.

Repository References

Key implementation files:

File	Role
`backend/app/simulation/core/model_definition.py`	Defines sets, parameters, variables, objective, and constraints
`backend/app/simulation/core/instance_builder.py`	Loads CSV sets and parameters into the Pyomo model
`backend/app/simulation/core/solver.py`	Maps solver names and runs HiGHS, Gurobi, or GLPK
`backend/app/simulation/README.md`	Documents the simulation workflow

External OSeMOSYS references:

- OSeMOSYS documentation: https://osemosys.readthedocs.io/en/latest/
- OSeMOSYS model structure: https://osemosys.readthedocs.io/en/latest/manual/Structure%20of%20OSeMOSYS.html
- OSeMOSYS project overview: https://osemosys.github.io/about/